Z Devops: Navigating your route to live with Git

Modernizing testing, promotion and deployment

Ian J Mitchell Dennis Behm

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0.1 Agenda

- Recap the basics of the recommended branching scheme
  - Focuses on the developer workflow
- "Environment branches" - yes, no or meh!
  - What's the point in hanging on to old practices?
  - It's just words... but words matter!
- The pipelines in action - demo time!

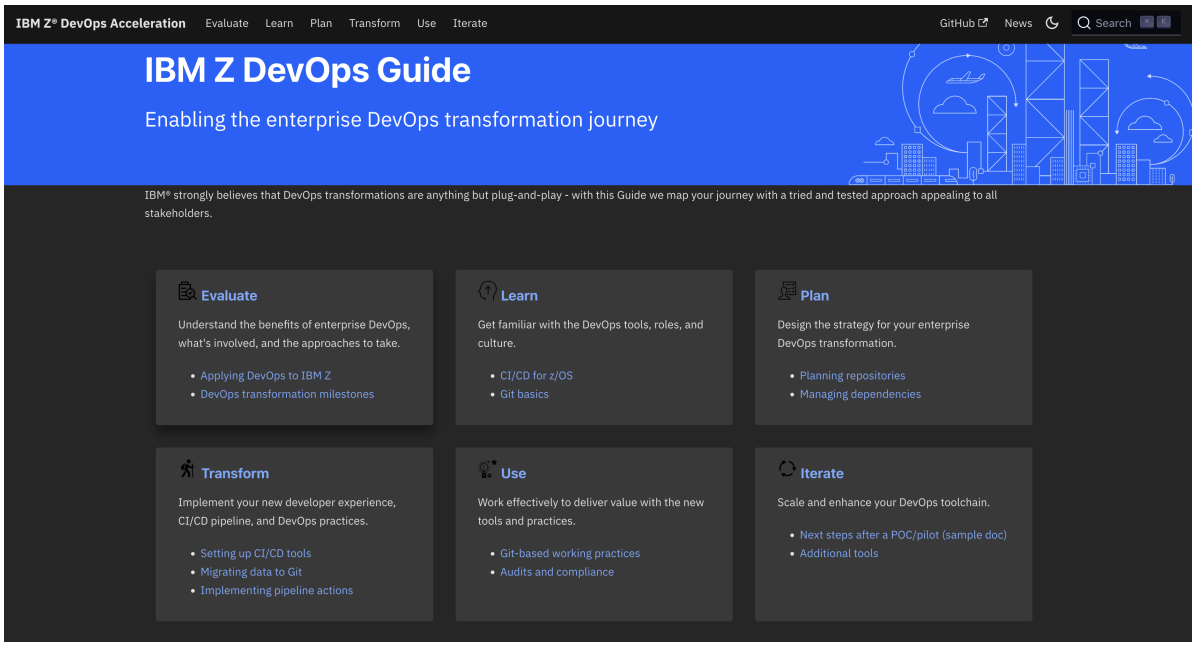
0.2 Setting the context

- Devops is foundational to modernization.
- Modernization of Devops is no longer a voyage of exploration.
  - (But plenty people will still try to convince you otherwise!)

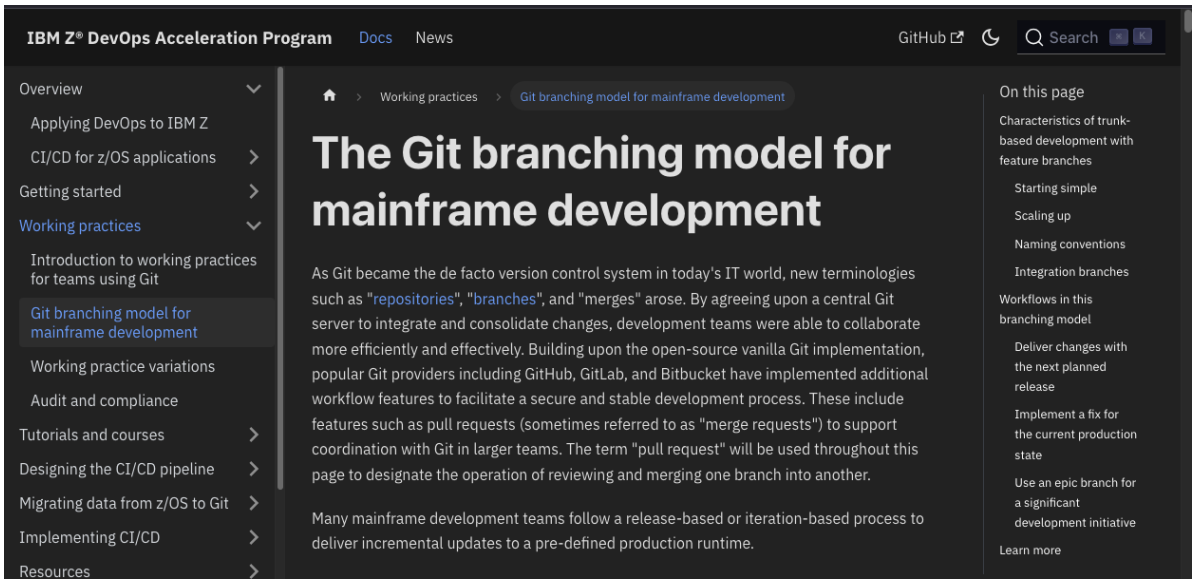
- No single tool or approach covers the entire journey to the destination.

We have a Guide... yes, it would be better if it were a map!

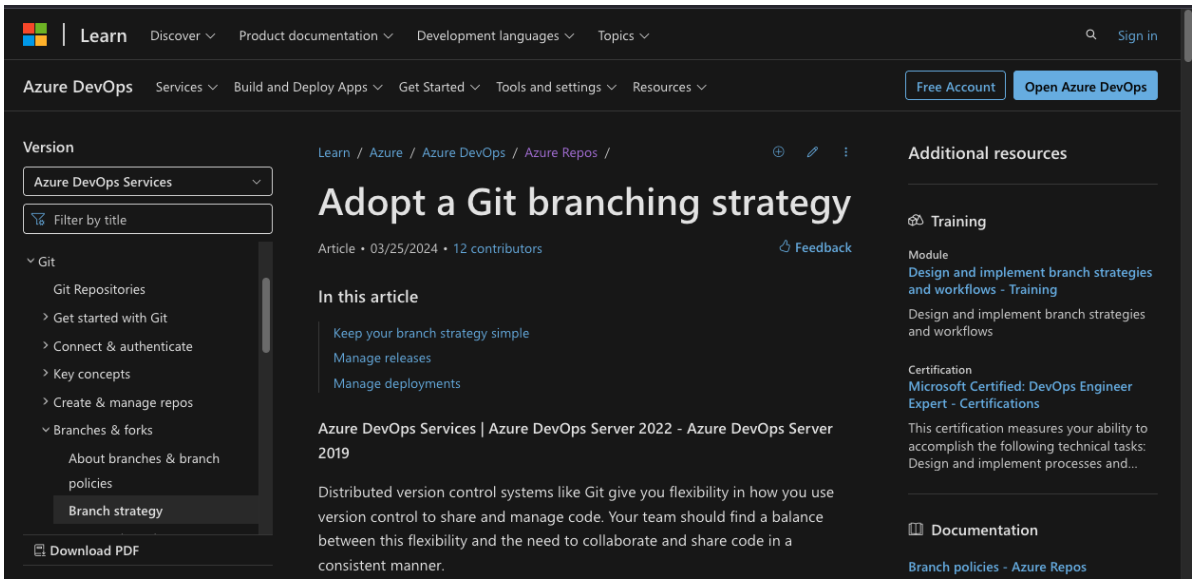
0.3 The Guide



0.4 Today's topic - Branching Strategy for Mainframe Teams



0.5 ... don't just take our word for it

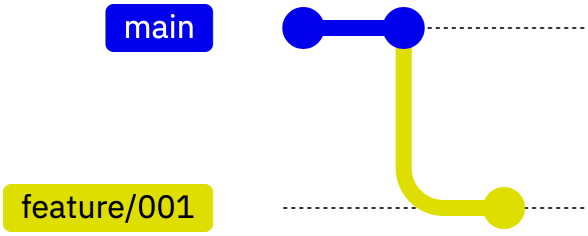


Microsoft Azure Devops Services

1 The Branching Strategy (Recap)

1.1 Starting Simple

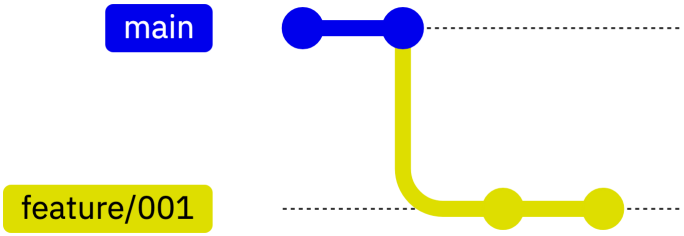
Every change starts in a branch.



- Developers work in the branch to make changes, perform user builds and unit tests.
- A branch holds multiple commits (changes to multiple files).

1.2 Starting Simple

Every change starts in a branch.



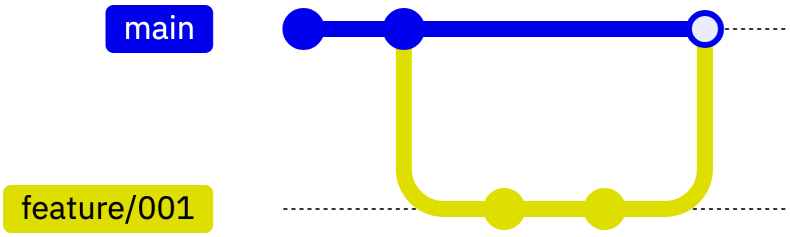
These changes on these branches are

- built,
- tested,
- reviewed and
- approved before merging to `main`.

1.3 Preparing to Merge into `main`

Feature Team/Developers will:

- Build
  - Builds may be done to any commit on any branch
  - Feature branch **must** build cleanly for a Pull Request
- Test
  - To prove quality of the changes in their feature branch



2 Environment branches - yes, no or meh!

2.1 The Route to Live

Comparing the recommended branching scheme to some common legacy patterns for workflows and build/deploy schemes

- These environments dominate the terminology and workflows with which the teams are familiar
- So there is an adjustment required to be fully comfortable with the recommended model

2.2 Branches for environments

Typically

- *DEV*
- *TEST*
- *PROD*

Let's look at the workflow for a set of features in a release.

2.3 Hanging on to legacy workflows

The following diagrams adopt the old names for:

- branches where developers work (‘*DEV*’),
- branches representing versions of the code base which are built and deployed to systems where formal tests are executed (‘*TEST*’) and
- branches representing the state of the deployed system in production (‘*PROD*’)

and shows associated build pipelines and systems on to which the built packages are deployed.

The pipelines targeting each environment may inject some characteristics specific to each environment.

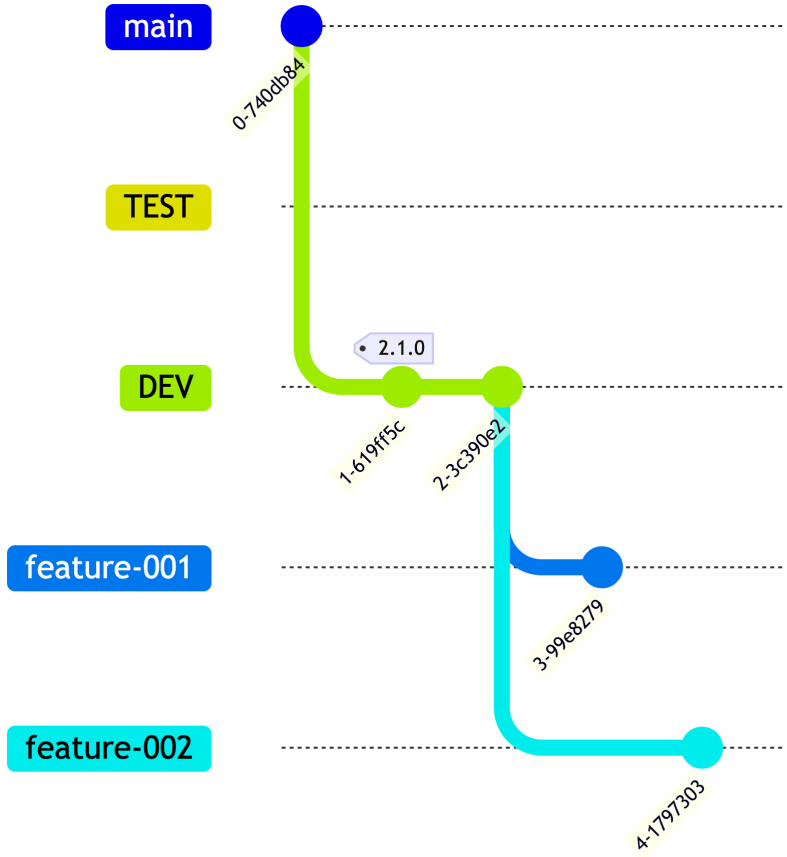
Obviously we maintain the principle that features will be implemented first in *DEV* and progress through one or more *TEST* branches and environments before deployment to *PROD*.

Equally obviously, *feature* branches are used by developers before they merge into *DEV*.

In order to be included in the pipeline build for any environment, the commits implementing a feature must be merged into the branch for that environment. Build˚vs are performed for any environment.

When one or more features are code-complete and merged into the *DEV* branch a new build is performed for deployment to the *DEV* environment.

2.4 Starting a development cycle

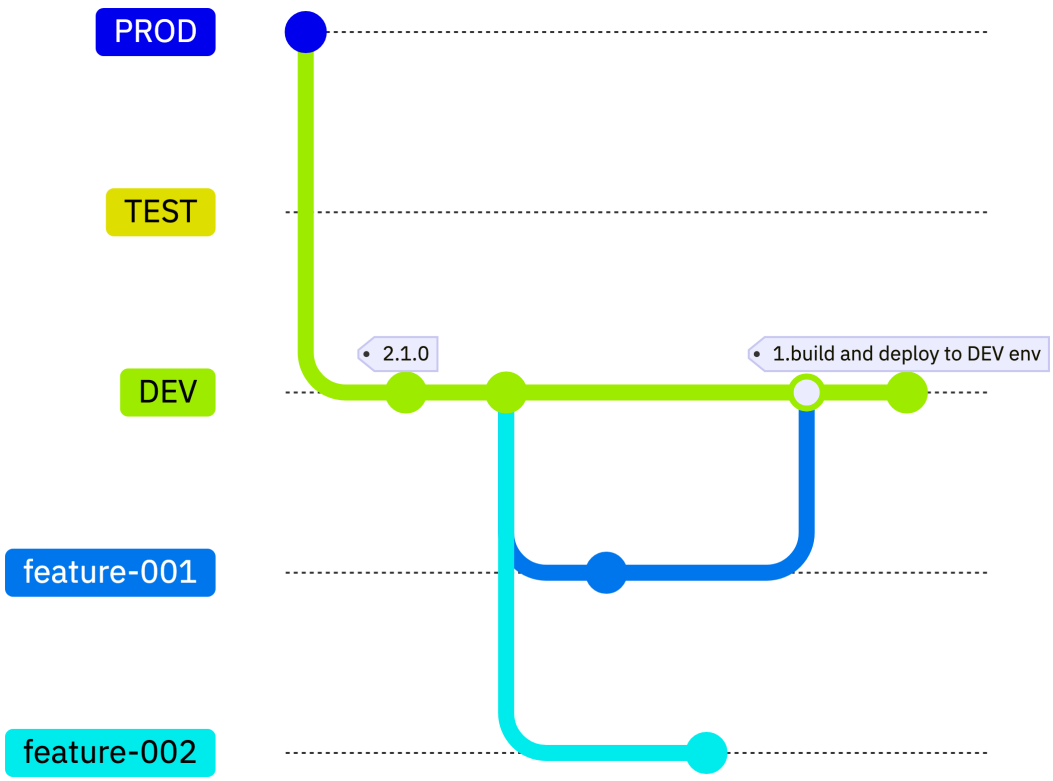


Both features for *release 2.1.0* are developed in their own dedicated branches.

2.5 Completing a feature

Feature-001 completes first and a PR is used to review and approve its merge into the *DEV* branch.

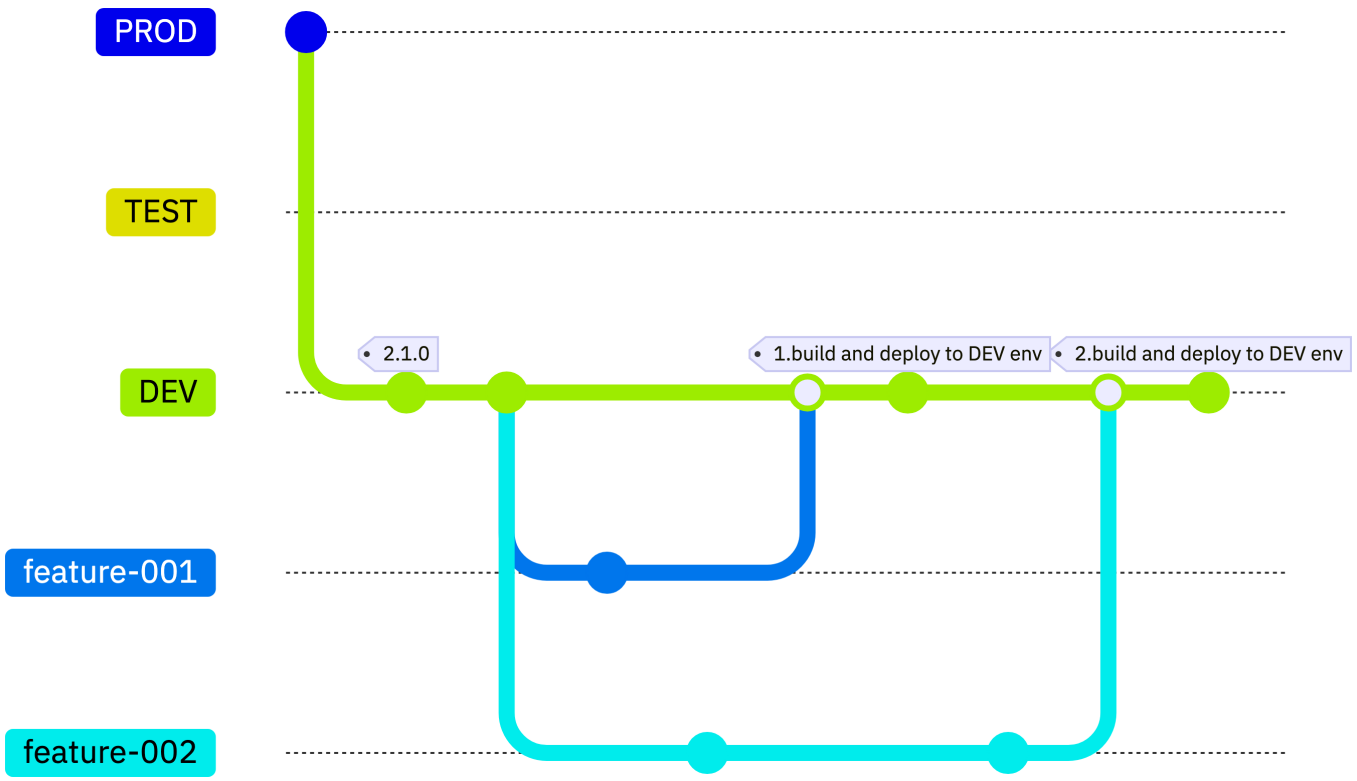
DEV is built, and the build is deployed to a *DEV* **environment**.



2.6 And another one...

The same workflow is performed when the second feature planned for the release is ready to be merged into *DEV*.

It's implicit that the feature-002 branch is synchronized with *DEV* after feature-001 is merged to *DEV* (at least before feature-002 can be merged with *DEV*).



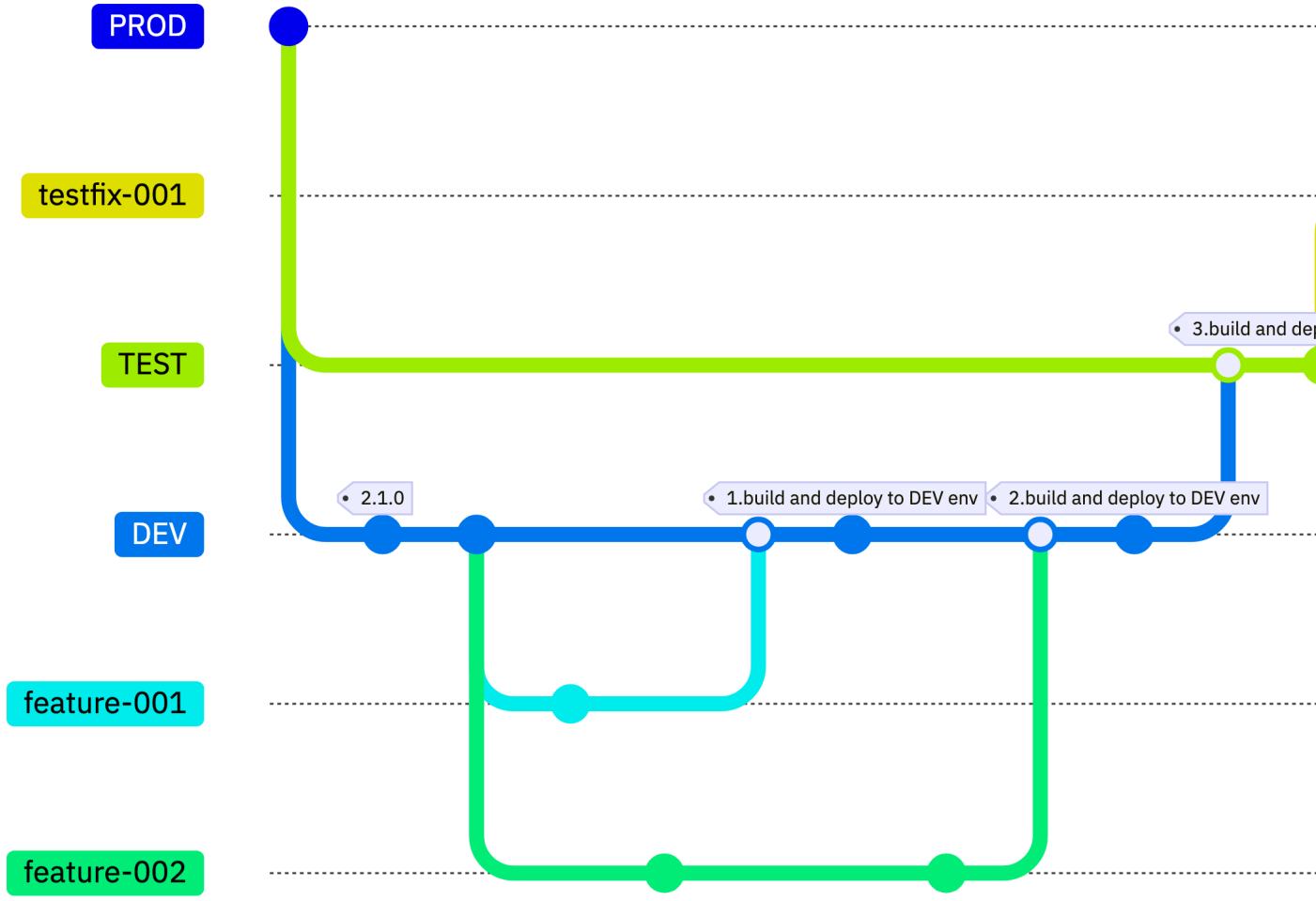
2.7 Testing features together

As these two features are now merged onto the *DEV* branch, they can be promoted together onto the *TEST* branch

- By performing a new combined build and deployment to the *TEST* **environment**.
- Here a problem is discovered and a fix to *TEST* is developed in a branch, merged into *TEST*, built and deployed for validation.

How can I test feature001 in the *TEST* environment without waiting on feature002 ? Or a different scenario: feature001 and feature002 are both in *DEV*, and the developer of feature001 wants to enter *TEST* already - how would I do this?

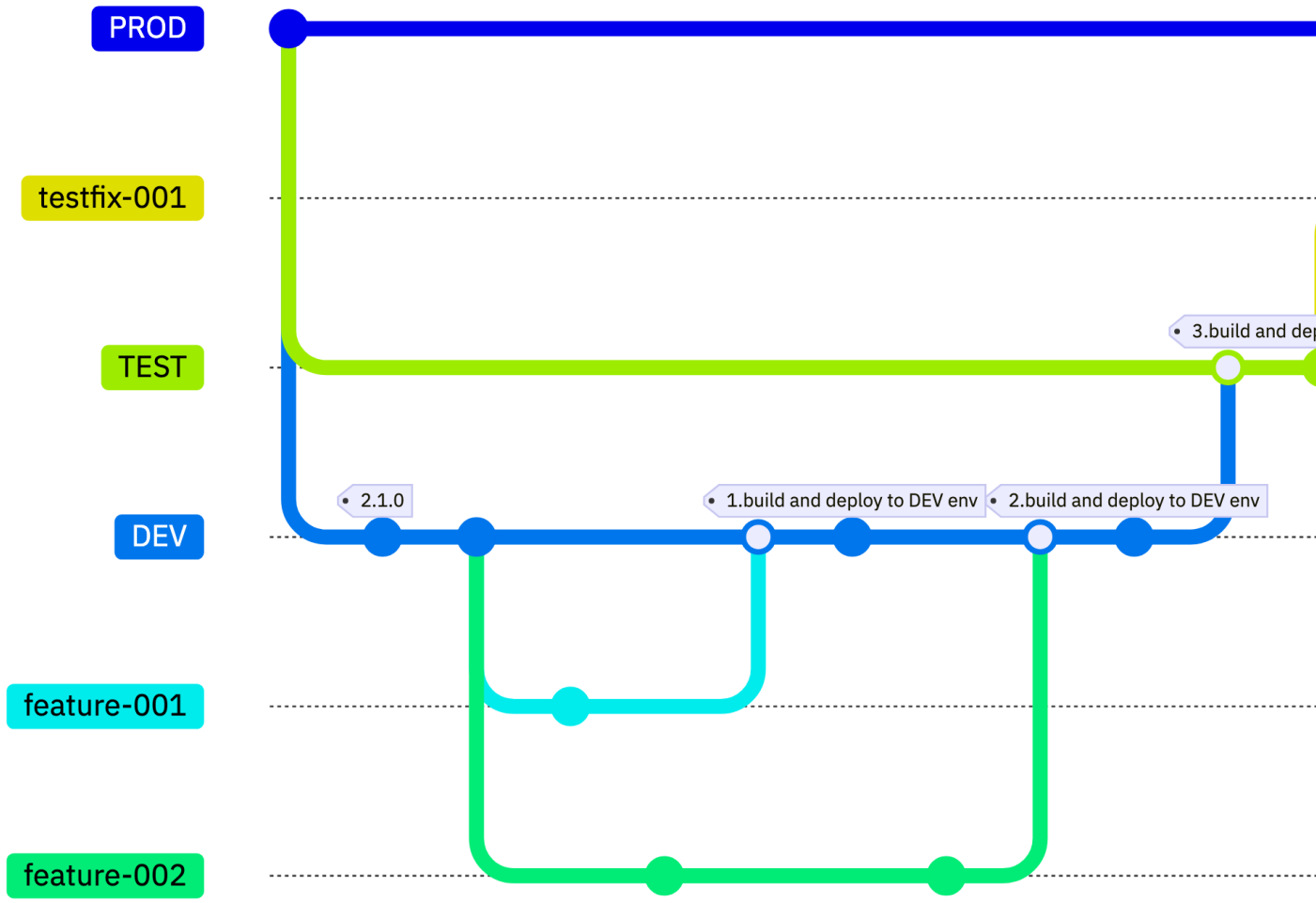
2.8 Testing and fixing problems



2.9 Deploying to production

After a successful test cycle, a PR is used to review and approve the merging of *TEST* (which now includes both planned new features) into the *PROD* branch.

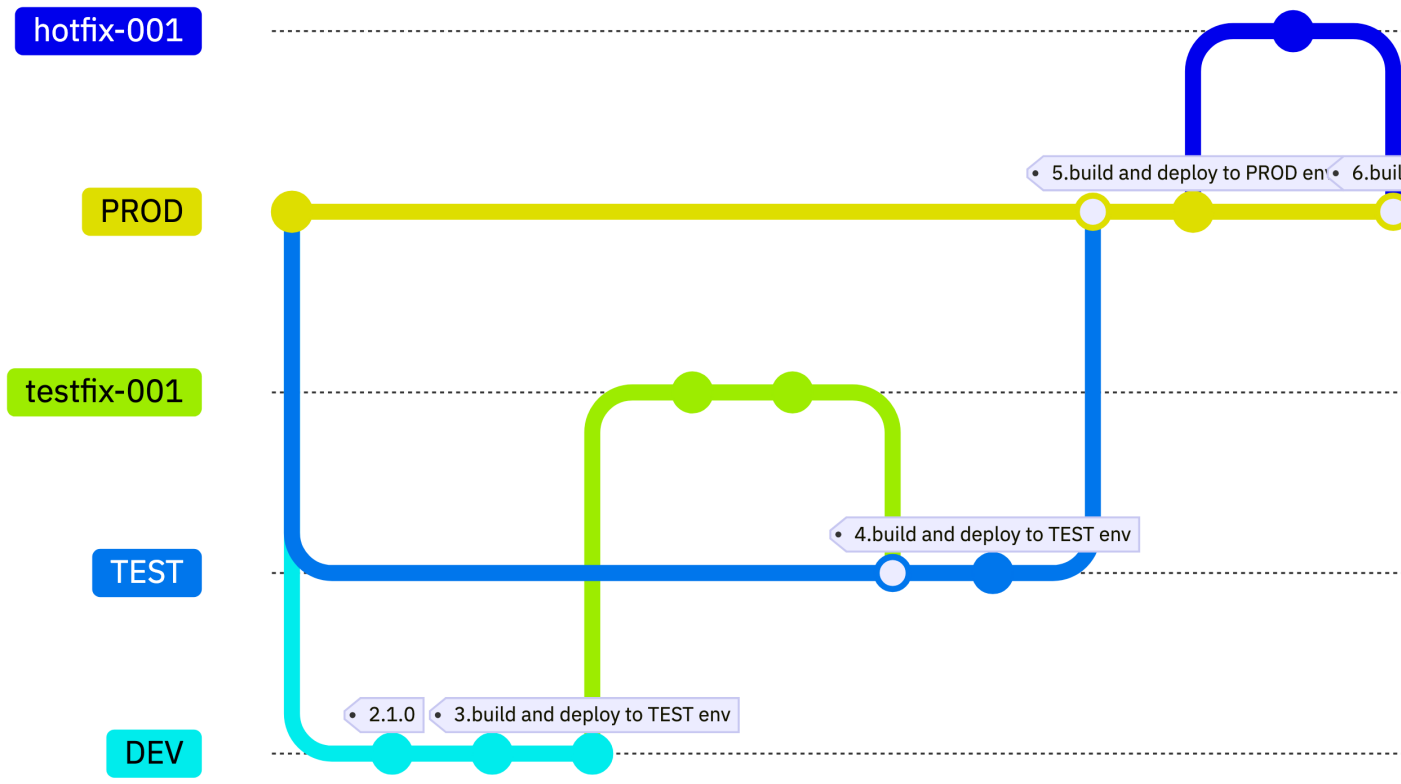
This is followed by a build and deployment to the *PROD* **environment**.



2.10 Fixing production problems

A HotFix is needed.

This is developed, reviewed and merged into the *PROD* branch with a new build and deployment to the *PROD* **environment** performed.



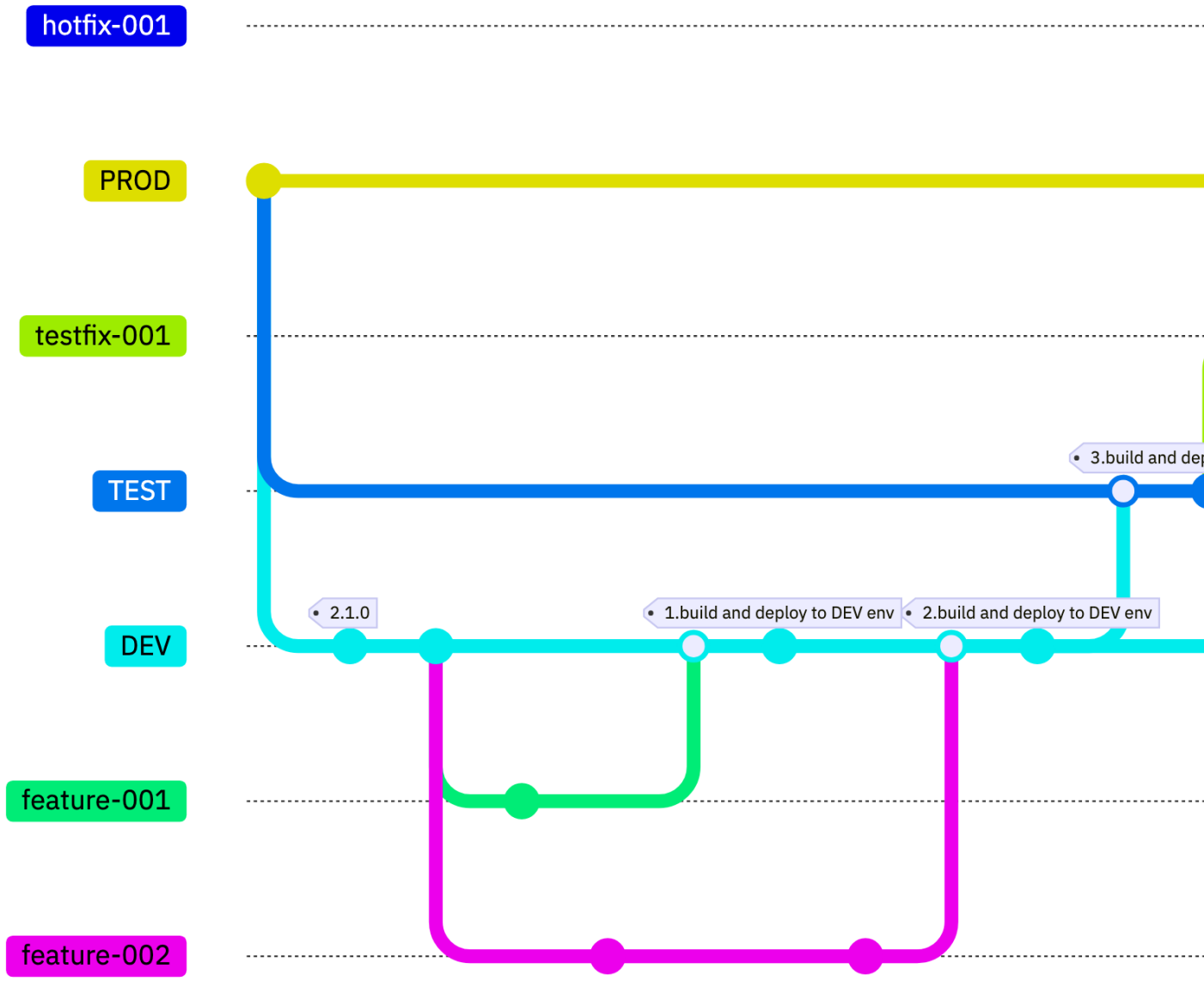
2.11 The next development cycle

Now the testing and release cycle for *release 2.1.0* is complete, both the *TEST* and *PROD* branches have fixes which need to be synchronized to *DEV*.

Since all the commits missing from *DEV* are present on *PROD*, we just `git merge` from *PROD* into *DEV*. And also from *PROD* into *TEST*.

And this is where you begin to ENTER the MERGE HELL

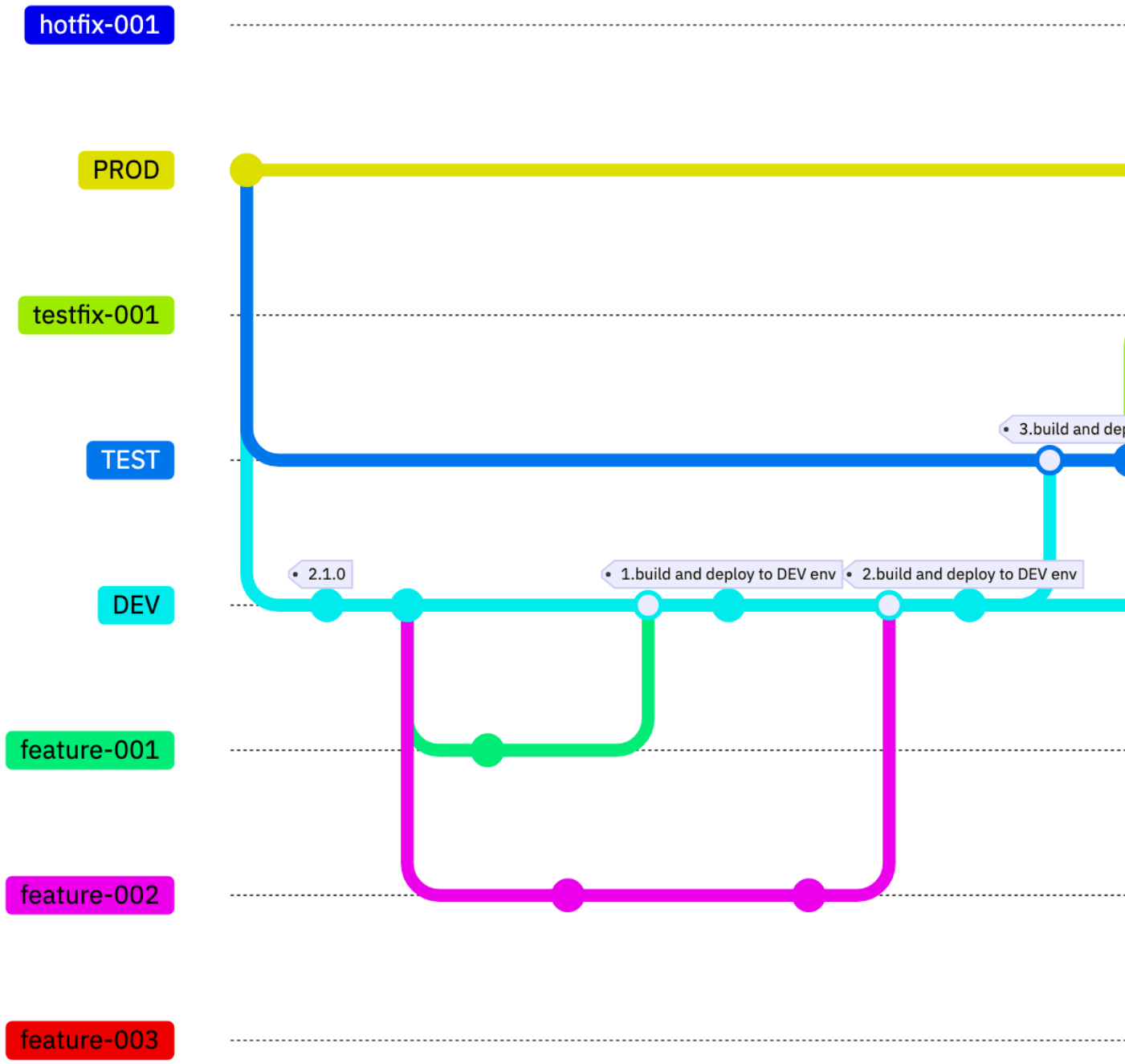
2.12



DEV is now identical to *PROD*.

Just as at the beginning of *release 2.1.0*, a new tag is used to demarcate the beginning of a new release cycle.

2.13



The cycle for *Release 2.2.0* goes smoothly with Feature-003 reviewed, approved, merged and tested into *DEV*, *TEST* and *PROD* with no need for any fixes!

(Which means there are no commits made to either *TEST* or *PROD* that need to go back to *DEV*.)

3 Comparing to the recommended model

3.1 Names, words and interpretation

The recommended model uses the industry-standard terms for branches in workflows reflecting common views of the roles they play.

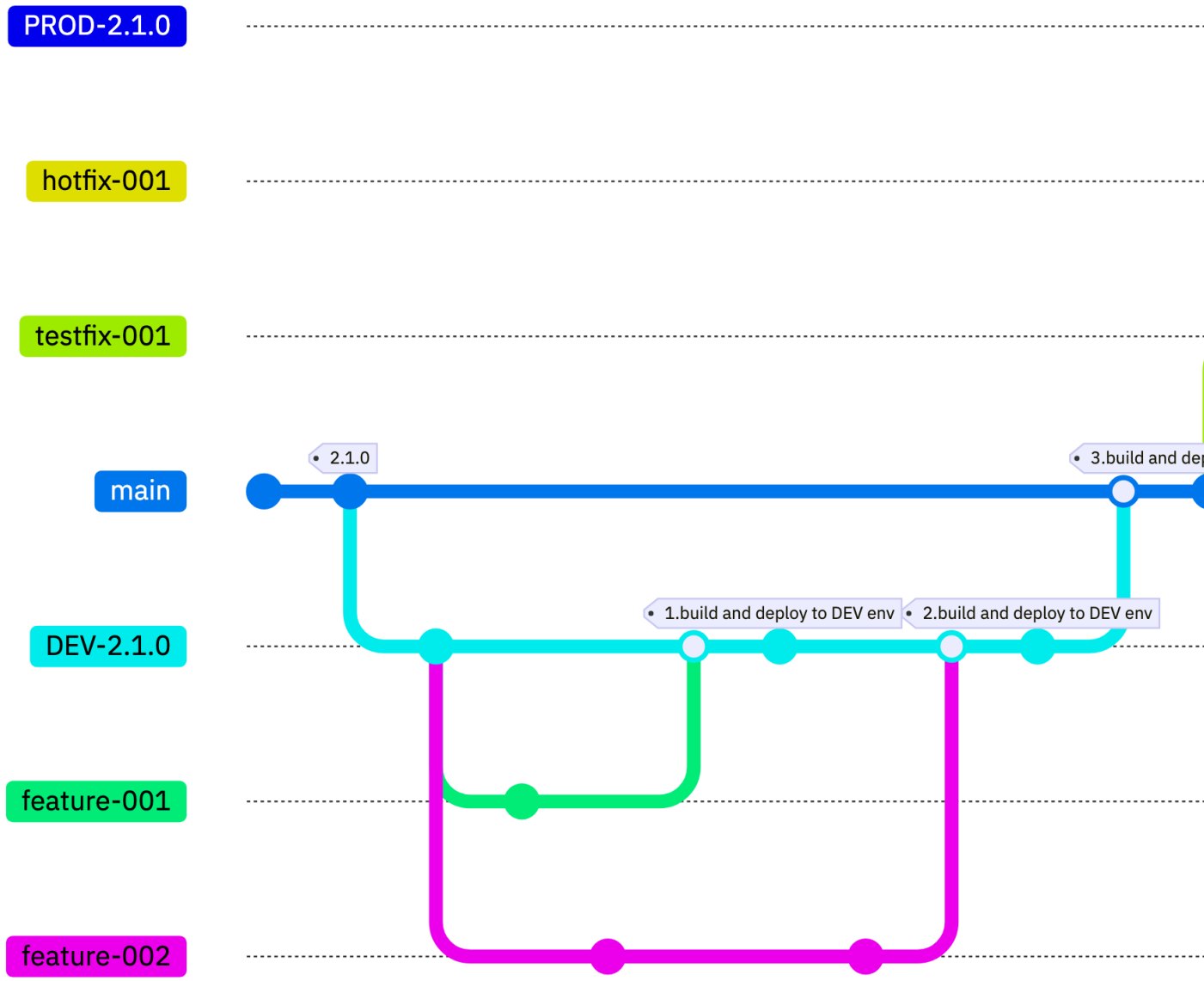
- Do these conflict with the legacy terms, or are they interpretations of the model achieving equivalent outcomes more easily?
- Is the legacy interpretation fundamentally different or close enough not to be a blocker on adoption?

3.2 Changing the vocabulary

The following diagrams show the same basic workflow actions and elements, but with the names from the recommended model.

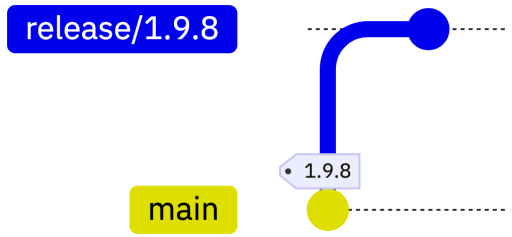
- A series of commits on `main` is equivalent to *DEV*
- A release build deployed into a test environment is equivalent to *TEST*
- Production versions (*PROD*) are demarcated by a Git tag on `main` - representing a series of releases builds that made it to production.

3.3


PROD-2.1.0
hotfix-001
testfix-001
main
DEV-2.1.0
feature-001
feature-002
2.1.0
3.build and dep
1.build and deploy to DEV env
2.build and deploy to DEV env

4 It's all the same!

4.1 Starting a development cycle

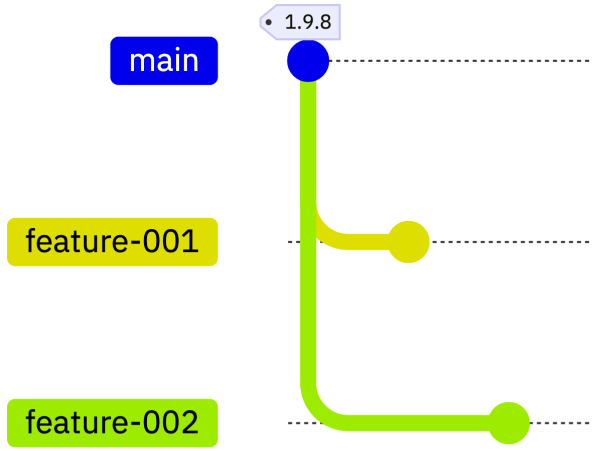


`main` is the most up to date branch at the beginning of every cycle. Release 1.9.8 is currently deployed to production via a previous cycle which built, tested and deployed it from the main branch.

A new `release/1.9.8` branch is created as an release maintenance branch to be the integration branch for production fixes.

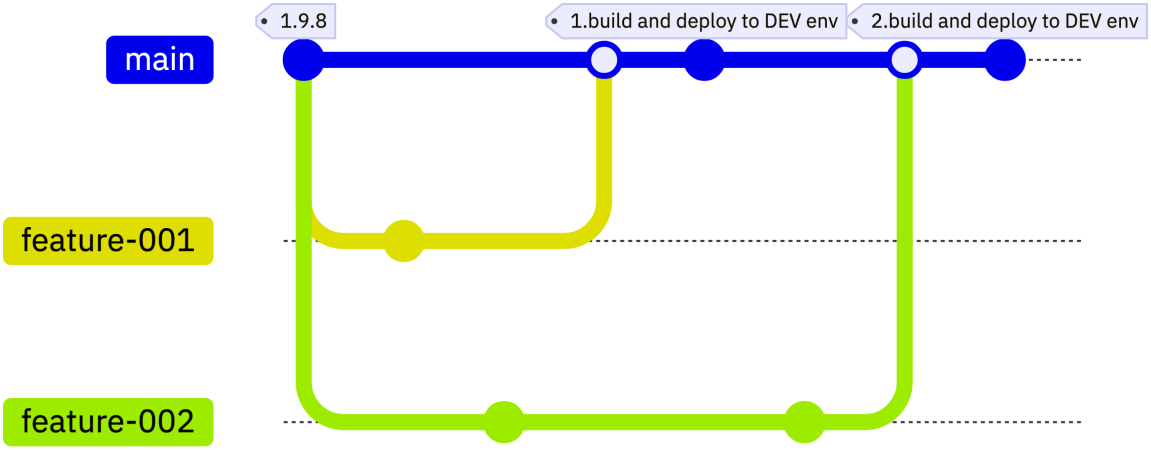
New features which will go into this next release cycle are added to main.

4.2 Features for the next release ...



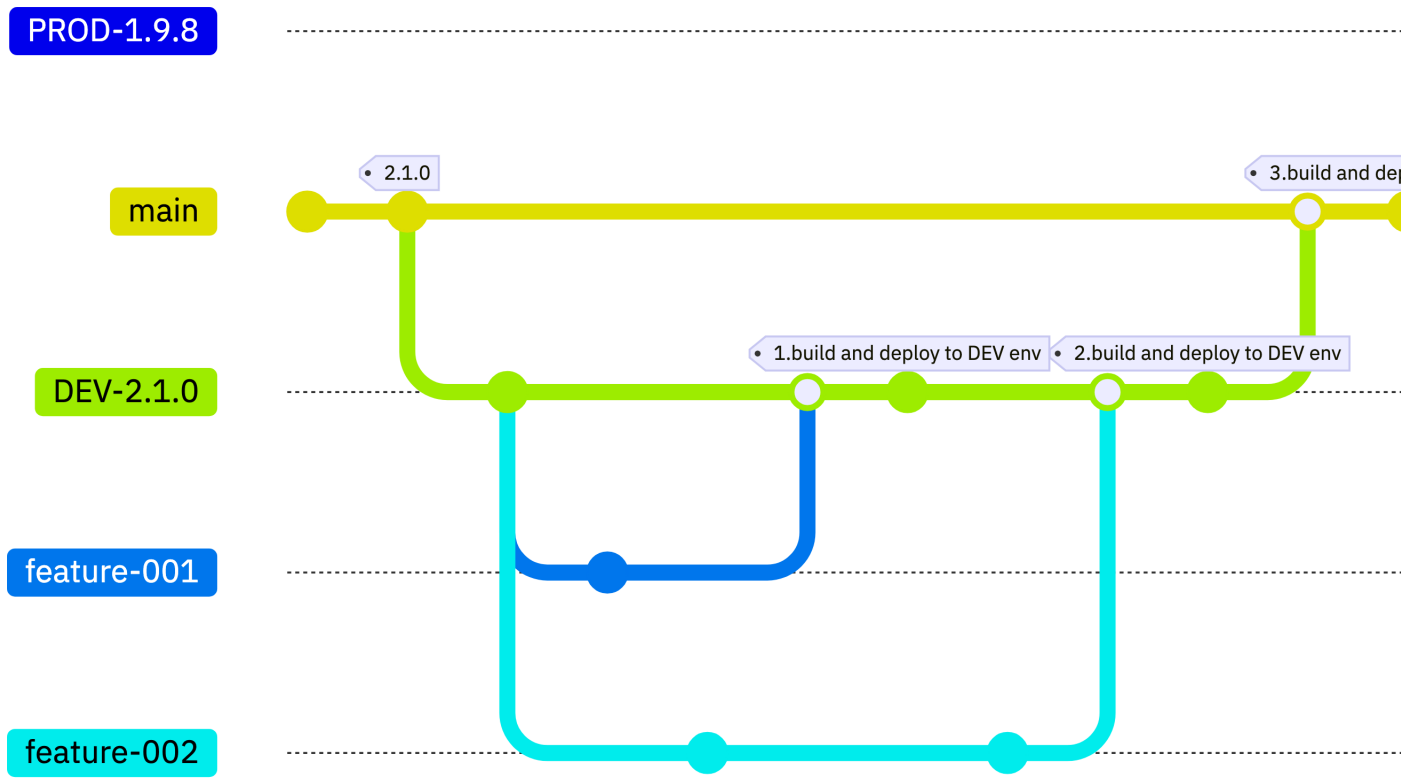
As usual, features are developed in their own branches.

4.3 Integrating features



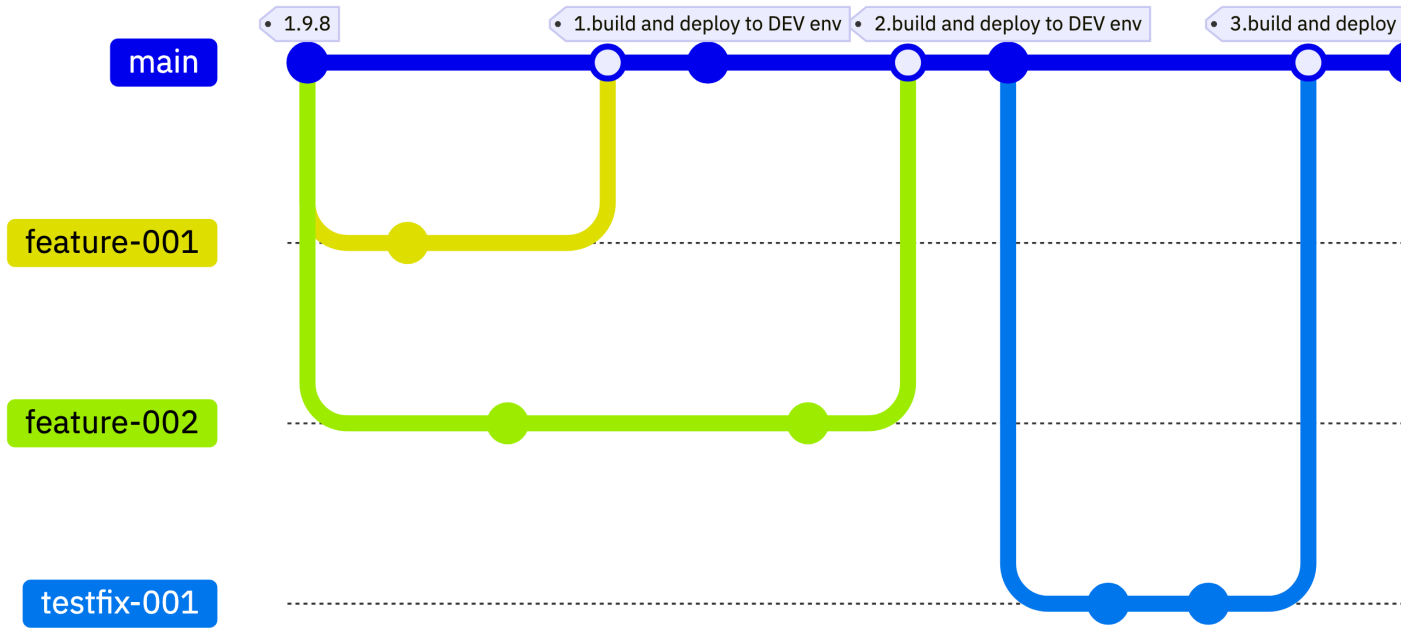
As each feature is completed, PRs are used to review, approve and merge them into the `main` branch where they can be tested together.

Again, it's implicit that the feature-002 branch is synchronized with *main* after feature-001 is merged to *main* (at least before feature-002 can be merged with *main*).



When both pass the tests for a DEV branch, a PR proposes a merge of them both to `main` where they can be reviewed and approved to be a release candidate.

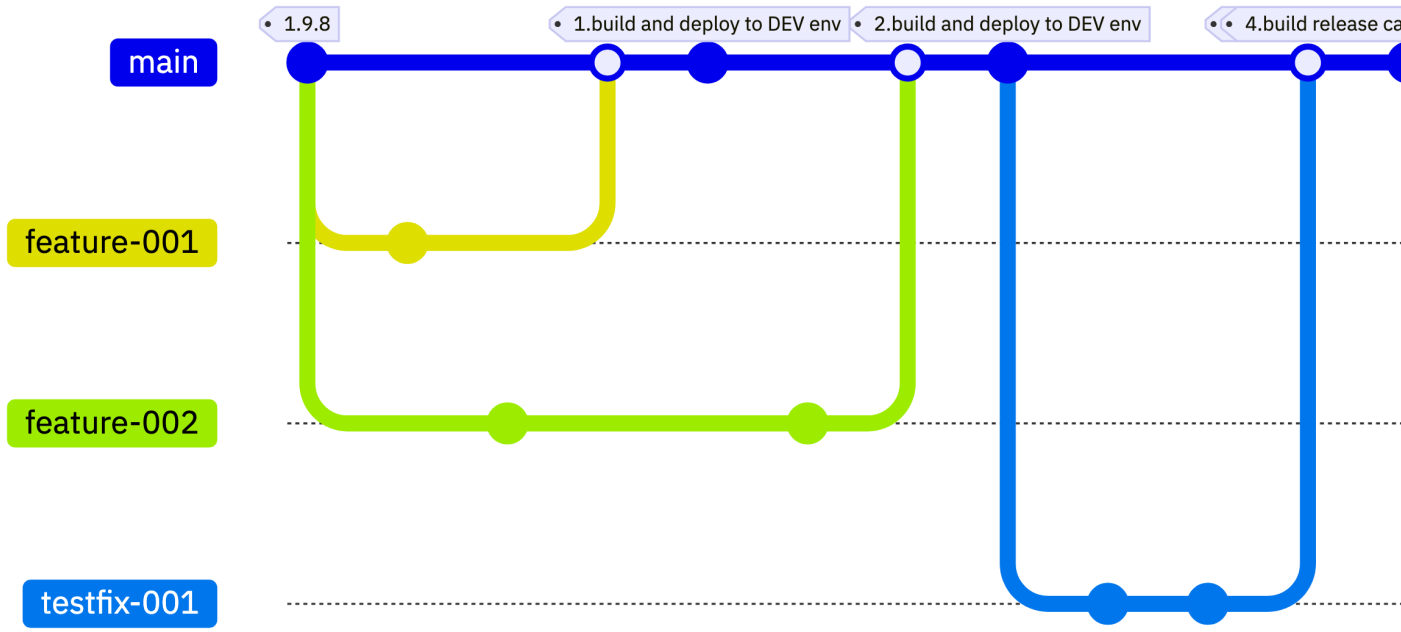
4.4 Testing and fixing problems ...



Again, the bug is discovered in testing, and `testfix-001` is required. It's developed in a branch, reviewed and approved and the build and testing refreshed.

4.5 Deploying to production

Now `main` has the new features and is passing the development tests, it's ready to consider a deployment to QA and then Production.

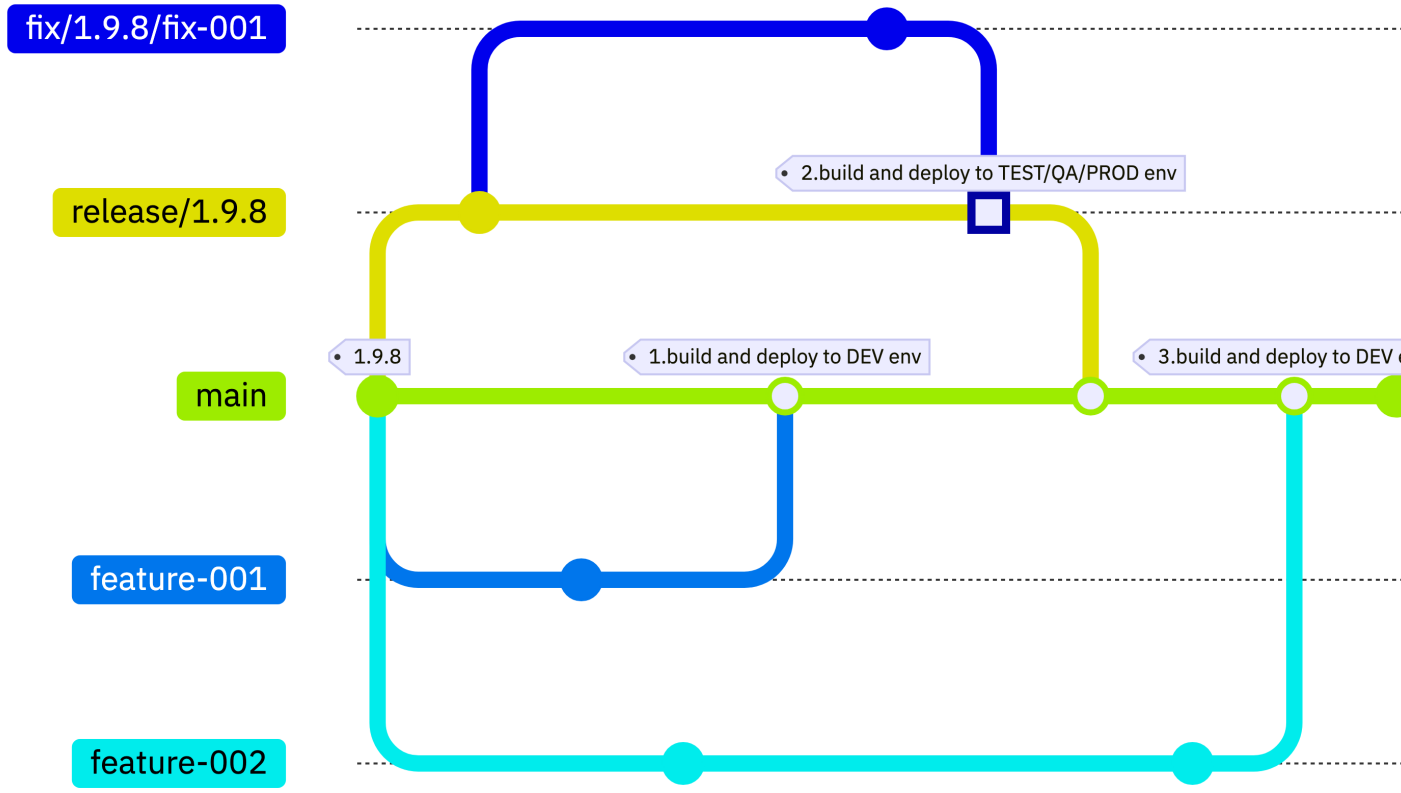


4.6 Fixing production problems

`main` has evolved, and a production issue is reported. What now?

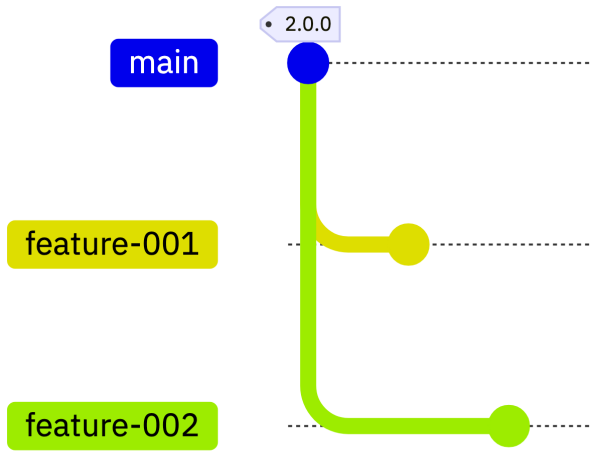
The bug which needs fixing is fixed with `fix-001` as a feature branch from the release maintenance branch `release/1.9.8`.

4.7



4.8 The next development cycle

To continue development for a *release 2.1.0*. Development of features on *DEV-2.2.0* proceeds in the usual way.



Build, testing and production deployment for next planned release 2.1.0 then proceeds as normal.

5 Advantages of the recommended model

5.1 `main` is the integration point

The *DEV* branch integrates multiple features before they are merged into *TEST* - which is the same role that the `main` branch has in the recommended model.

It signals that the target integration branch for developers is functionally equivalent to a series of merged *feature* branches.

5.2 Equivalence

Since *DEV* in the legacy and `main` in the recommended model would both be the target of merging new changes, it makes **no functional difference** whether:

- a set of **long-lived** branches are maintained and continuously need to be refreshed, a.k.a rebase *DEV* with the `HEAD` of *TEST* if *TEST* has fixes merged on to it OR
- a series of merged feature branches representing the changes for the next release.

It makes NO DIFFERENCE to the history or the validity of the `main` branch or the feature branches developers work off if there's a single long-lived branch synchronized to the HEAD of *TEST* or if a new feature branch is created for fixing code for the next planned release from `main`.

5.3 `main` is *TEST*

TEST is the integration branch for all new features - essentially identical to the branch we call `main` in the recommended model, containing the series of new changes that are planned for the upcoming release.

- In legacy terms, *DEV* would advance ahead of *TEST* with feature branches being merged into it exactly as feature branches would be ahead of `main` in the recommended model.
  - As commits from feature branches are merged into *DEV*, the *DEV* pipeline builds and deploys testable packages to the *DEV* environment. Having proven their goodness, a PR is created to merge *DEV* into *TEST*.

    These additional are not required in the recommended model.

If nothing else has merged into *TEST* since the last PR from *DEV* then this is straightforward. But if it's permitted for developers to merge commits for fixes into *TEST* without merging them into *DEV* first, then *DEV* and *TEST* will have diverged and *DEV* will need to be brought up-to-date with the commits from *TEST* which are missing.

Commits normally merge according to the upward arrows in both diagrams, but there are a number of downward pointing arrows - these signify synchronization points where a branch which is normally a source of commits to another (*DEV* to *TEST*, *TEST* to *PROD*) need to incorporate commits which were made independently to the (normally) target branch.

This is exactly as `main` and *feature branches* are used. Thus *TEST* is playing the role we assign to `main` in the recommended model. It seems entirely reasonable that the branch to which developers aim to merge their work and which is the main stage of testing has the conventional common name of `main`.

5.4 Release maintenance branches are identical to *PROD*

The legacy interpretation demands that commits for new features are merged from *TEST* to *PROD*, but also that it is allowed for *hot-fixes* to be merged with *PROD* in emergencies.

- *TEST* cannot be behind *PROD* (ie missing any hot-fixes) when a new set of commits are to be released to *PROD*.
- Taking the commits from *TEST* which are newer than the last to *PROD* makes their histories identical.

This makes it functionally the same as creating git tags to label commits on `main` for release builds and create a release maintenance branch when required in the recommended model.

6 Summary

6.1 Summary

- Legacy terminology is a blocker for alignment to common ways of working
  - and makes on-boarding harder
- Our recommended model decouples the state of the code from the environments
  - for which the code is built
  - and where it is deployed

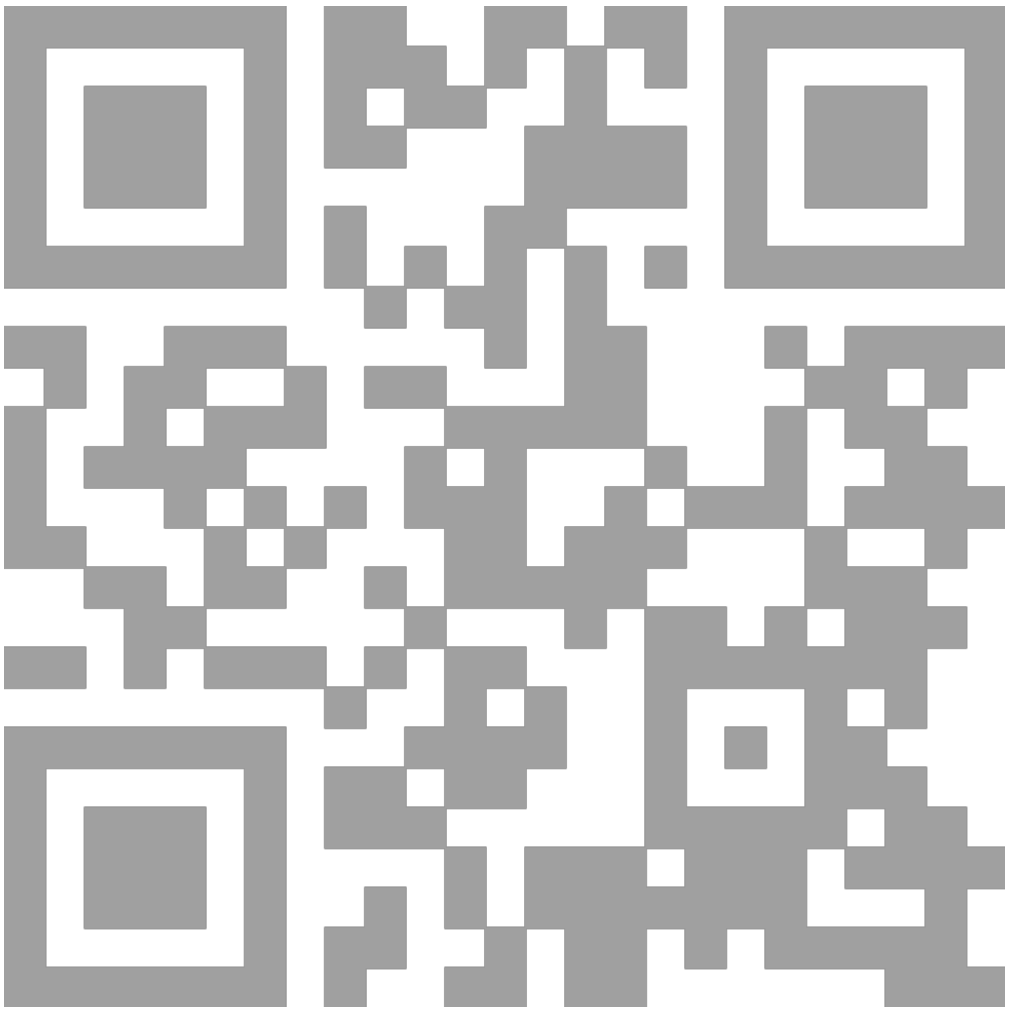
Words...

- *DEV* is equivalent to a series of changes
  - oriented to projects/features
- *TEST* is equivalent to `main`
  - so let's use the standard term
- *PROD* is equivalent to a series of release builds marked by tags deployed to production
  - all with the history and traceability we all need

7 THANK YOU!

7.1 Session Feedback

- Submit your feedback using the Whoova app or at the conference website - https://conferences.gse.org.uk/2025/feedback/dh
  - (*Make sure you are signed into Whoova app or MyGSE if using the website*)
- The session code is **DH**



7.2 IBM Z Day


//* IBM Z Day
//* open | on | secure
12 November 2025
IBM TechXchange Virtual Event
IBM Z's largest tech event of the year
IBM Z Makes More Possible – More Precision | Security | Innovation
Don't miss IBM Z Day 2025 on November 12th from 8 AM – 4:30 PM ET (IBM Z Skills Acceleration track starts at 12 AM ET), for an exclusive look at the latest innovations from IBM Z and LinuxONE. Earn 4 industry recognized badges and take your career to new heights.
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165+ COUNTRIES
LEARNING PLATFORM FOR ALL
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App Dev & IT Ops
Security
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Extended 16 ½ hour track (12 AM to 4:30 PM ET)
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7.3 Mainframe Playlist